
Priority-Conditioned Retention: A Controllable Middle-Layer Memory Circuit in Pretrained Mamba

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Abstract

State-space models (SSMs) such as Mamba compress an entire history into a fixed-size continuous hidden state, achieving linear-time inference at the cost of associative recall: unlike attention, they cannot losslessly retrieve an arbitrary past token. Prior work attributes this to a capacity bound but leaves open whether the practical failure is a *storage* limit or a *control* limit. We show that a pretrained Mamba-790m already contains a controllable, spatially localized memory circuit, accessible at inference with no fine-tuning. By scaling the pre-softplus input to the Δ gate at positions downstream of a target, we reduce the effective decay applied to that target and recover associative recall on a capacity-pressured multi-query associative-recall (MQAR) task ($0.53 \rightarrow 0.73$ at the optimal operating point). The effect is dose-dependent, target-specific, and bidirectional, and it is localized to layers 16–23. Restricting the intervention to these 8 of 48 layers recovers most of the recall gain at near-zero language-modeling cost (perplexity $\times 1.07$), whereas the all-layer intervention inflates perplexity $\times 1.81$. We frame this as the continuous-state analogue of importance-aware KV-cache compression and report an honest boundary: a literal write-quantization analogue does *not* transfer, because the superimposed state has no per-item slots to reallocate.

1 Introduction

Mamba and related selective SSMs match Transformers on many language tasks while running in linear time, but remain markedly weaker at *associative recall* — retrieving a specific value bound to a key seen earlier in context. Jelassi et al. (2024) show this stems from a fundamental capacity constraint: a fixed state cannot losslessly copy arbitrarily long inputs. Yet a capacity bound does not, by itself, explain *behavioral* recall failure at moderate load. The failure could be (a) a **control** problem — the recurrence applies a default recency decay and does not preferentially keep query-relevant items — or (b) a **storage** problem — the state is genuinely out of bits. These have opposite implications: (a) is fixable at inference; (b) is not.

This paper provides evidence for (a) in a regime where it matters. We do *not* modify or retrain the model. Instead we ask: does a pretrained Mamba already expose a handle that lets us trade its default recency policy for a target-preserving one? We find that it does, that the handle is the Δ gate, and — most importantly — that the handle is localized to a contiguous band of middle layers. Operating only on that band recovers recall essentially for free in terms of language-modeling quality, the behavior one expects of a dedicated mechanism rather than a diffuse one.

Contributions.

- A test-time intervention on the Δ gate that controllably preserves a target item in a capacity-pressured SSM, with dose-response, specificity, and bidirectionality controls (Section 4.1).
- Localization of this control to layers 16–23 of Mamba-790m, and a *surgical* result: the 8-layer intervention matches most of the full-model recall gain at perplexity $\times 1.07$ vs. $\times 1.81$ (Section 4.2). This is our central finding.
- A framing placing this control on a single axis with importance-aware KV-cache compression, plus an honest negative result: a literal write-quantization analogue fails, bounding how far the cache analogy carries (Section 5).

We report scope honestly: all positive results are for Mamba-790m on synthetic MQAR; we observe non-replication on Mamba-1.4b and fragility on natural language (Section 6).

2 Background and Framing

The Δ gate as a retention knob. In a selective SSM the state updates as $h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t$, with $\bar{A}_t = \exp(\Delta_t A)$ and $A < 0$, so $\Delta_t > 0 \Rightarrow \bar{A}_t \in (0, 1)$. The contribution of token i surviving to a query at position T is therefore $\bar{B}_i x_i \cdot \prod_{\tau>i} \bar{A}_\tau$. Two facts follow. First, the default policy is recency-biased: older contributions decay geometrically. Second, the *survival* of an item is governed by the *downstream* decays $\prod_{\tau>i} \bar{A}_\tau$, not by the item’s own write alone. Reducing Δ at positions after a target pushes those $\bar{A}_\tau \rightarrow 1$, slowing the decay the target experiences — a preservation lever requiring no weight change.

Discrete $\leftrightarrow$ continuous importance allocation. The KV-cache compression literature (Liu et al., 2024; Hooper et al., 2024; He et al., 2024; Zandieh et al., 2026) implements one principle: spend more bits on more important tokens. This presumes a slotted cache with random access. Mamba has no slots — all items are superimposed in one vector. We therefore treat “importance $\rightarrow$ retained information” as architecture-agnostic and instantiate it in Mamba not as per-token bit allocation but as *differential retention* through the Δ gate. We name this framing Priority-Conditioned Retention (PCR). Section 5 reports where the analogy holds and breaks.

3 Method

Task. Multi-query associative recall (MQAR) with random single-token key/value words (no semantic priors). A prompt presents N key–value pairs followed by a query key; the model must emit the bound value. Capacity pressure is induced by increasing N ; we query the first pair as the default target. As N grows the base model degrades smoothly (recall $0.77 \rightarrow 0.53 \rightarrow 0.33$ for $N=16, 32, 64$); we adopt $N=32$ (base recall 0.53) as the operating point that balances pressure and headroom.

Intervention. We register forward hooks on each layer’s `dt_proj` and multiply its pre-softplus output by a per-position gain g . Setting $g < 1$ (“other-gain”) at positions downstream of the target’s value reduces their effective decay. A layer mask restricts the intervention to a chosen subset of layers. The model runs on its pure-PyTorch sequential path (fused kernels intentionally not installed) so the Δ pathway is hookable. No parameters are updated.

Model and metrics. `state-spaces/mamba-790m-hf` (48 layers, fp32). We report top-1 recall over 30 random seeds per condition, and perplexity on a fixed held-out passage under the same gain, relative to the unmodified model.

4 Results

4.1 Test-time control

Sweeping other-gain at $N=32$ over all layers yields a clear inverted-U with an optimum at $g=0.8$, where recall rises $0.53 \rightarrow 0.73$, then collapses for $g \leq 0.6$ as integration halts (Figure 1). The non-monotonicity is mechanistically expected: lowering Δ both slows decay ($\bar{A}_\tau \rightarrow 1$, helpful) and shrinks the write $\bar{B}_\tau = \Delta_\tau B_\tau$ (harmful); past $g \approx 0.7$ the latter dominates and integration stalls.

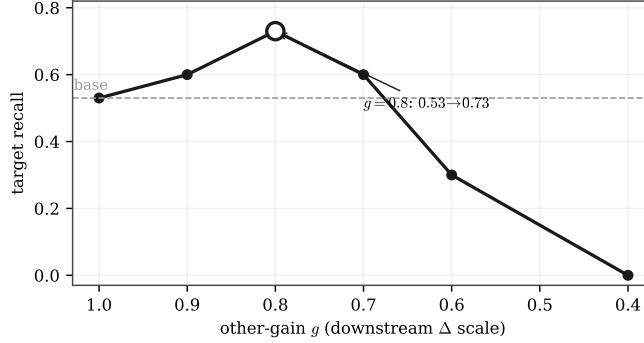


Figure 1: Dose–response of the Δ -gate intervention (Mamba-790m, MQAR $N=32$). Recall peaks at $g=0.8$ ($0.53 \rightarrow 0.73$) and collapses when Δ is over-suppressed, confirming Δ as a graded retention knob.

Table 1: Specificity and direction controls at $N=32$, $g=0.8$ (base recall 0.53). Only targeted downstream suppression helps.

Condition	Target recall
Base (no intervention)	0.53
Downstream suppression ($g=0.8$)	0.73
Random-position suppression ($g=0.8$)	0.50
Upstream / increase Δ	< 0.53

The effect is **target-specific**: applying $g=0.8$ to downstream positions yields 0.73, whereas the same gain budget on *random* positions yields only 0.50 (near base), ruling out a generic “more gain helps” artifact (Table 1). It is also **bidirectional**: *increasing* Δ degrades recall below base, so the sign of the intervention maps to the sign of the effect.

4.2 Layer localization and the surgical circuit

Restricting $g=0.8$ to 8-layer bands isolates the responsible depth: recovery concentrates in **L16–23** (+0.13), with a secondary contribution at L24–31 (+0.07) and nothing elsewhere (Figure 2). Crucially, the localization is also *cheaper*: Table 2 shows the 8-layer intervention captures most of the recall gain at near-zero language-modeling cost, while the all-layer version pays a large perplexity tax. A targeted edit to the memory circuit is almost free; a global one is not — the signature of a dedicated mechanism rather than a diffuse property.

5 The Discrete–Continuous Boundary

If retention is one lever, *precision* — the literal KV-quant analogue — is a candidate second lever: quantize the write of low-priority tokens to fewer bits. We tested this by quantizing the per-position convolution output (the token’s written representation) to k bits. Single-item precision is not a graceful knob: target recall is flat from 16 $\rightarrow$ 8 bits, then cliffs (8-bit 0.53 $\rightarrow$ 4-bit 0.03 $\rightarrow$ 0). Reallocation fails outright: quantizing *distractor* positions to 4 bits drives target recall to 0.00 (Figure 3, Table 3) — it corrupts the shared computation rather than freeing capacity.

This is informative, not merely a null. Because the state is a superposition with no random-access slots, lowering one item’s write precision either does nothing (≥ 8 -bit) or injects noise the whole sequence shares (≤ 4 -bit). There is no per-item bit budget to reallocate. The importance-allocation principle thus transfers to Mamba *only* through retention (Section 4.2), not through quantization — a concrete boundary on the KV-cache analogy and an empirical instance of the “no slots” property that distinguishes continuous-state memory from a discrete cache.

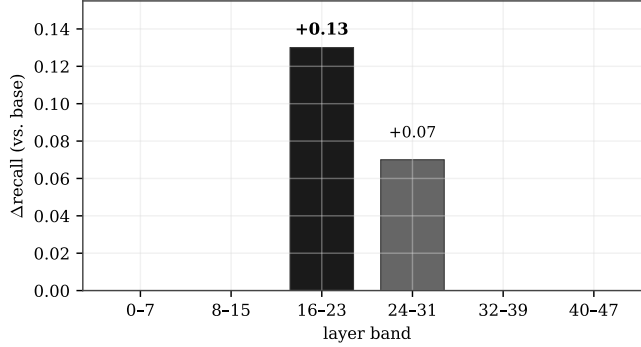


Figure 2: Recovery localizes to a contiguous middle band. Applying $g=0.8$ to one 8-layer band at a time, the recall gain concentrates in L16–23 (+0.13), with a smaller effect at L24–31; other depths contribute nothing.

Table 2: Test-time intervention on Mamba-790m, MQAR $N=32$ (base recall 0.53, base ppl ≈ 232). The 8-layer edit captures most of the gain at a fraction of the perplexity cost.

Intervention	Recall	Perplexity
All layers, $g=0.8$	0.73	$\times 1.81$
L16–23 only, $g=0.8$	0.67	$\times 1.07$

6 Limitations and Scope

We state these plainly; they bound the claim and we believe reporting them strengthens it. **Single model:** all positive results are for Mamba-790m; on Mamba-1.4b the same intervention shows no positive circuit at a comparable pressure regime (all layer-band $\Delta \leq 0$ despite ample headroom), so the L16–23 circuit appears model-specific and we do not claim a universal depth. **Synthetic task:** effects are robust on random-token MQAR but fragile on natural-language multi-fact recall, where the intervention can hurt. **Graded priority is not established:** a three-level graded variant appeared monotone but is confounded with query position; we do not claim multi-level allocation. The SSM constraint ($\Delta > 0 \Rightarrow \bar{A} < 1$, shared decay) predicts that truly item-selective allocation is limited, consistent with Section 5.

7 Related Work

SSM capacity. Jelassi et al. (2024) establish the copying/capacity limit; work on primacy/recency documents the default decay policy. We add a controllability result on top of the capacity picture.

KV-cache compression. KIVI, KVQuant, ZipCache and rotation-based quantizers (Zandieh et al., 2026) realize importance-aware bit allocation on slotted caches; we ask whether the principle survives the move to a slotless continuous state, and find it survives as retention but not as quantization.

Recall and the “what to keep” frontier. Prompting methods such as Just Read Twice (Arora et al., 2024) and architectures such as Based, DeltaNet and Titans (Behrouz et al., 2024) target the same problem, mostly by training new architectures or memory modules; ours is a training-free, frozen-base alternative.

Mechanistic interpretability. Localizing a behavior to a contiguous layer band parallels circuit-style findings in Transformers; ours is, to our knowledge, an early such localization of a *controllable* memory mechanism in a pretrained SSM.

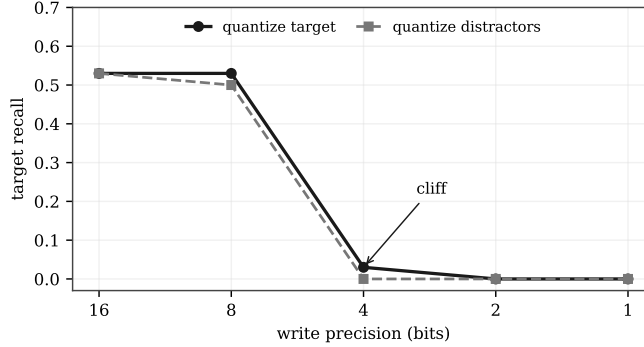


Figure 3: Write-quantization is not a graceful lever. Lowering a single item’s write precision either does nothing (≥ 8 bits) or collapses recall (≤ 4 bits); quantizing distractors to free capacity instead destroys the target, evidencing the absence of per-item slots.

Table 3: Write-precision quantization at $N=16$. Neither targeting nor reallocation behaves like bit allocation in a slotted cache.

Quantize	16-bit	8-bit	4-bit	2-bit	1-bit
target	0.53	0.53	0.03	0.00	0.00
distractors	0.53	0.50	0.00	0.00	0.00

8 Conclusion and Future Work

A pretrained Mamba’s associative recall is, in part, a controllable property of a specific middle-layer band, not a fixed ceiling. A surgical, training-free intervention on layers 16–23 recovers recall at negligible language-modeling cost. The importance-aware compression principle transfers to continuous state through retention but not through quantization. Our proposed next step is to remove the oracle: a small *question-conditioned, auto-localizing* controller trained on a frozen base, evaluated against Belady-MIN retention efficiency under streaming queries, toward a training-free improvement of the recall–capacity frontier of pretrained SSMs.

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